

Problem 4. Let V_n be the vector space of all polynomials of degree at most n in a variable t , so V_3 contains things like $2t^3 - t^2 + 5t + 1$. In this problem and the next, we will think of polynomials as inputs and outputs to functions, rather than functions themselves.

(a) Let $M : V_1 \rightarrow V_2$ be the function $M(f(t)) = (t-1)f(t)$. Check that M is a linear transformation, and find the matrix representing it with respect to the bases $1, t$ for V_1 and $1, t, t^2$ for V_2 .

(b) Let $S : V_3 \rightarrow V_3$ be the function $S(f(t)) = f(t-1)$, so for example

$$S(t^2 + t) = (t-1)^2 + (t-1) = t^2 - t$$

This is a linear transformation. Find the matrix representing S with respect to the basis $1, t, t^2, t^3$ for V_3 . Is this transformation invertible?

Solution. (a) To check that M is linear, we need to check that for $f, g \in V_1$, $c \in \mathbb{R}$, we have the equations

$$M(f+g) = M(f) + M(g)$$

$$M(c \cdot f) = c \cdot M(f)$$

The first equation comes out to

$$(t-1)(f(t) + g(t)) = (t-1)f(t) + (t-1)g(t)$$

which is an instance of the distributive law for polynomials. The second comes out to

$$(t-1)(c \cdot f(t)) = c(t-1)f(t)$$

which is also true. To find the matrix representing M with respect to bases $1, t$ for V_1 and $1, t, t^2$ for V_2 , we find where M sends the basis for V_1 :

$$M(1) = t - 1$$

$$M(t) = t^2 - t$$

We then express these elements of V_2 in terms of the basis for V_2 .

$$t - 1 \rightsquigarrow \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$t^2 - t \rightsquigarrow \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

These coordinate vectors are the columns of the matrix:

$$\begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

(b) As in part (a), to find the matrix representing S , we see where it sends the basis elements $1, t, t^2, t^3$, then find the coordinates of the resulting polynomials with respect to the basis:

$$S(1) = 1 \rightsquigarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$S(t) = t - 1 \rightsquigarrow \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$S(t^2) = (t - 1)^2 = t^2 - 2t + 1 \rightsquigarrow \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

$$S(t^3) = (t - 1)^3 = t^3 - 3t^2 + 3t - 1 \rightsquigarrow \begin{bmatrix} -1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$$

These vectors are the columns of the matrix representing S :

$$\begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The transformation S is indeed invertible. There are a few ways of seeing this. One is to observe that the matrix representing S can be transformed into the identity matrix by row operations, so is invertible. The inverse matrix then corresponds to a transformation $V_3 \rightarrow V_3$ which is an inverse of S . Another way to see that S is invertible is to describe the inverse explicitly:

$$S^{-1}(f(t)) = f(t + 1)$$